MASTER'S THESIS PLAN

A Hierarchical ReAct Multi-Agent Pipeline for Residential Appliance Load Scheduling Under Dynamic Electricity Pricing

Student	**Nguyen Han Hoan Le**
Degree programme	Information Processing Science, University of Oulu
Main supervisor	Yueqiang Xu
Scope	Master's thesis, 30 ECTS
Methodology	Design Science Research (DSR)
Planned period	1 October 2026 to 31 March 2027
Goal completion date	31 March 2027 (thesis submitted for review in Laturi)
Language	English

1. Background and motivation

Households on dynamic electricity contracts can lower their bills by running flexible appliances, such as washing machines, dishwashers, tumble dryers and electric vehicle chargers, in low-price hours. In Finland more than a third of households had a dynamic-price contract in 2024 (Energy Authority, 2025), and the day-ahead market moved to fifteen-minute prices in October 2025, which makes the price curve finer and more volatile. Coordinating several appliances under such a curve is difficult without automation, because the household must also keep its total load below a peak limit set by its main fuse or by a power-based network tariff. Home energy management systems that schedule appliances in this way have long been studied, with mixed-integer linear programming as a standard way to find the optimal schedule (Beaudin & Zareipour, 2015; Sou et al., 2011).

Households can already install rule-based automations, for example Home Assistant blueprints that switch a device on during the cheapest hours (Kert, 2023) or planners that choose the cheapest block of hours (Daniel, 2022), and optimisers such as EMHASS (EMHASS contributors, 2026). Rules handle each appliance on its own and ignore the household's total load, while optimisers require the household to describe every appliance in structured form and do not explain their decisions. Large language models (LLMs) have been used as natural-language interfaces to such systems (Michelon et al., 2025) and as agents combined with optimisation (Raghavan & Giridhar, 2025). Agents that reason and act with tools, following the ReAct pattern (Yao et al., 2023), offer both a natural-language interface and explanations. First agentic home energy management systems have been proposed (El Makroum et al., 2026; Jonah et al., 2026), but they were evaluated on few days, did not compare against the tools households already use, and reported coordination failures when several appliances compete for the same cheap hours.

2. Research gap and objectives

Based on a preliminary review, few studies appear to evaluate an LLM-based multi-agent home energy management system across different price patterns, household loads and peak limits, and against the home energy management tools that households already use. This will be confirmed by a structured literature search at the start of the work. The objectives of the thesis are:

- **Design:** a hierarchical ReAct multi-agent system that turns a household's daily appliance plan into a reliable, cost-saving recommendation with a short explanation.
- **Reliability:** every recommendation must respect each appliance's time window and the household peak limit; a verification step returns flawed schedules to the agents for revision.
- **Evaluation:** a reproducible benchmark built from measured appliance cycles and real Finnish prices, with the optimal schedule as a reference, and a pre-registered confirmatory test.
- **Practicality:** a response time and an API cost per recommendation that suit daily use.

3. Research questions

RQ: How can a hierarchical ReAct multi-agent system be designed to give households reliable, cost-saving daily appliance recommendations under dynamic electricity prices?

SQ1 (value and cost): How much do the pipeline's recommendations save, how often are they valid, and what do they cost to produce, compared with a language model that answers without tools?

SQ2 (conditions): Under which price patterns, household loads and peak limits do the recommendations save more than, or as much as, existing home energy management tools available to households, namely cheapest-hours automations and the EMHASS optimiser?

The primary outcome is the saving against starting every appliance as soon as it is ready. The following hypotheses will be registered before the confirmatory evaluation; the exact thresholds will be fixed together with the supervisor.

	Hypothesis	Planned test
H1a	The pipeline saves more than the LLM without tools.	Wilcoxon signed-rank
H1b	Its recommendations are valid more often than those of the LLM without tools.	Wilcoxon signed-rank
H1c	Its mean response time is below 60 s and its mean API cost below USD 0.10 per recommendation.	One-sided t test against the limit
H2a	The pipeline saves more than the cheapest-window rule.	Wilcoxon signed-rank
H2b	It saves more than the cheapest-hours automation.	Wilcoxon signed-rank
H2c	It saves the same as EMHASS within one percentage point.	Paired equivalence test (TOST; Lakens, 2017)

4. Research methodology

The thesis follows Design Science Research (Hevner et al., 2004), structured by the six-phase process of Peffers et al. (2007): the central activity is building and evaluating an artifact rather

than observing an existing phenomenon. The evaluation strategy follows the FEDS framework (Venable et al., 2016): artificial, simulated household days, formative evaluations during the design iterations and a summative evaluation at the end.

DSR phase	Activity in this thesis	Output
1 Problem identification	Structured literature search in Scopus and IEEE Xplore, reported with a PRISMA 2020 flow diagram (Page et al., 2021)	Research gap; confirmed research questions; literature review chapter
2 Objectives	Define success criteria: saving, validity, response time and API cost; hypotheses	Objectives and evaluation design
3 Design and development	LangGraph pipeline with an Orchestrator, an Appliance Profiler, a Price-Aware Scheduler and a Conflict Resolver; shared tools; verification step; reference systems	Working artifact; design chapter
4 Demonstration	Run the pipeline on pilot scenarios and inspect its traces	Demonstration results
5 Evaluation	Exploratory benchmark for design iterations, then a pre-registered confirmatory run on held-out scenarios	Results and statistical tests
6 Communication	Thesis, design guidelines, open code and data, interactive demo	Thesis submitted; demo

4.1 Artifact

The artifact will be built in Python with LangGraph. Each specialist agent follows the ReAct Thought-Action-Observation loop with only the tools of its role, and the agents pass their results through a shared state. The Orchestrator reads the household's request, the Appliance Profiler describes the appliances, the Price-Aware Scheduler plans the start times, and the Conflict Resolver removes peak-load conflicts; this division of roles follows work on LLM multi-agent systems (Guo et al., 2024; Hong et al., 2024). A deterministic verification step checks every schedule and returns flawed ones for revision. Candidate extensions to be tested as design iterations are worked examples of similar solved days, retrieved as in retrieval-augmented generation (Lewis et al., 2020), and an optimiser tool that the agents call, check and explain, as in work that pairs language models with solvers (Li et al., 2023). The primary model is GPT-4o-mini; a small open-weight model run locally will be tested as a secondary option.

4.2 Evaluation

- **Benchmark:** household days that combine measured appliance cycles from the REFIT dataset (Murray et al., 2017) with Finnish Nord Pool day-ahead prices (Elering AS, 2026), in a factorial design of four price patterns (stable, volatile, price spike, negative prices), two household loads and two peak limits.
- **Two sets:** an exploratory set for development and design iterations, and a held-out set with fifteen-minute prices for the confirmatory test, run only after the hypotheses are registered.
- **Compared systems:** the pipeline; the same LLM without tools; the Home Assistant cheapest-hours automation; a cheapest-window rule; EMHASS run as deployed; and the

mathematically optimal schedule (mixed-integer linear programming; Sou et al., 2011) as an upper bound. A simple rule written for the study serves as a further reference point.

- **Measures:** saving against no scheduling, share of valid schedules, gap to the optimum, response time, tokens and API cost; three runs per scenario for the LLM systems.
- **Analysis:** paired non-parametric tests and an equivalence test as listed above, with effect sizes and a descriptive breakdown by price pattern, load and peak limit.

5. Data and research data management

- **Data sources:** the REFIT electrical load dataset (CC BY 4.0) and public Nord Pool day-ahead prices for Finland, retrieved through the Sähkötin and Elering data services and checked against ENTSO-E. All sources are cited in the thesis.
- **Personal data:** none. REFIT is anonymised household data, and no human participants are involved. Only benchmark data, never personal data, is sent to the language model API.
- **Storage:** code, scenarios and results are kept in a version-controlled Git repository with regular backups. Raw REFIT and price files are not redistributed; scripts download and rebuild them. API keys are kept in a local configuration file outside version control.
- **Reproducibility:** fixed random seeds, a frozen benchmark, logged runs and traces, and automated tests for the scoring and the reference systems.
- **Sharing and retention:** after approval the code, benchmark and anonymised results will be made openly available alongside the thesis, which will be published open access.

6. Ethics and responsible use of AI

The study involves no human participants and no personal data, so no ethical review is required. Generative AI tools may assist with programming and with searching the literature; any such use will follow the University of Oulu guidelines and be stated in a declaration in the thesis. The language models evaluated in the study are objects of research, not authors of the text.

7. Expected results and contribution

- **A designed and evaluated artifact:** an open-source hierarchical ReAct pipeline for daily appliance recommendations.
- Empirical evidence on what the recommendations save, how reliable they are and what they cost, compared with an LLM without tools and with the tools households already use (SQ1, SQ2).
- A reusable benchmark and evaluation protocol for LLM-based home energy management.
- Design guidelines for multi-agent systems in which a language model must produce reliable, constraint-respecting decisions.

8. Schedule

The work is planned for about 26 weeks of full-time study, from 1 October 2026 to 31 March 2027, with a two-week break over the holidays. Supervision meetings are planned every two weeks.

Period	DSR phase	Main tasks	Milestone
1 Oct to 1 Nov 2026 (weeks 1 to 5)	1 Problem identification	Structured literature search and screening; research gap; confirm research questions with the supervisor	Research plan approved in Laturi; literature review outline
2 Nov to 29 Nov 2026 (weeks 6 to 9)	2 Objectives	Success criteria and hypotheses; data download; build the benchmark; optimal reference and rule-based systems	Benchmark and reference results; methodology chapter draft
30 Nov 2026 to 17 Jan 2027 (weeks 10 to 16, break 21 Dec to 3 Jan)	3 Design and development	Implement the agents, tools and verification step; LLM without tools; EMHASS setup; pilot tests	Working pipeline; design chapter draft
18 Jan to 7 Feb 2027 (weeks 17 to 19)	4 Demonstration and formative evaluation	Demonstration on pilot days; exploratory benchmark; design iterations; register the hypotheses	Final design frozen; pre-registration
8 Feb to 28 Feb 2027 (weeks 20 to 22)	5 Summative evaluation	Confirmatory run on the held-out set; statistical tests; condition breakdown	Results chapter draft
1 Mar to 21 Mar 2027 (weeks 23 to 25)	6 Communication	Discussion and conclusions; abstract; complete draft to the supervisor; interactive demo	Full draft for supervisor feedback
22 Mar to 31 Mar 2027 (weeks 26 to 27)	6 Communication	Revision after feedback; language check; Oulu formatting; originality check	Thesis submitted for review in Laturi by 31 Mar 2027

9. Planned structure of the thesis

- **1 Introduction:** motivation, research problem and questions, DSR approach.
- **2 Literature review:** home energy management, LLM-based agents for energy, ReAct, multi-agent LLM systems, retrieval and solver tools, the Finnish electricity market, research gap.
- **3 Research methodology:** DSR process, literature search, theoretical foundations, evaluation strategy, benchmark and data.
- **4 System design and implementation:** architecture, agents, tools, verification step, design iterations.
- **5 Evaluation and results:** demonstration, exploratory results, confirmatory tests, conditions.
- **6 Discussion:** answers to SQ1 and SQ2, implications, limitations, validity.
- **7 Conclusions:** answer to the RQ, contributions and future work.

Key references

Beaudin, M., & Zareipour, H. (2015). Home energy management systems: A review of modelling and complexity. Renewable and Sustainable Energy Reviews, 45, 318–335. https://doi.org/10.1016/j.rser.2015.01.046

Daniel. (2022). Nordpool Planner [Home Assistant custom component]. GitHub. Retrieved 5 October 2026, from https://github.com/dala318/nordpool_planner

El Makroum, R., Zwickl-Bernhard, S., & Kranzl, L. (2026). Agentic AI home energy management system: A large language model framework for residential load scheduling. Results in Engineering, 29, 109857. https://doi.org/10.1016/j.rineng.2026.109857

Elering AS. (2026). Nord Pool day-ahead prices, bidding zone FI [Data service; data owner Nord Pool]. Retrieved 26 September 2026, from https://dashboard.elering.ee/api/nps/price

EMHASS contributors. (2026). EMHASS: Energy Management for Home Assistant (Version 0.18.4) [Computer software]. https://emhass.readthedocs.io

Energy Authority (Energiavirasto). (2025, July 10). Dynamic-price contracts held by more than one third of households: Fixed-term contracts still the most common type in 2024 [Press release]. https://energiavirasto.fi/en/-/dynamic-price-contracts-held-by-more-than-one-third-of-households-fixed-term-contracts-still-the-most-common-type-in-2024

Guo, T., Chen, X., Wang, Y., Chang, R., Pei, S., Chawla, N. V., Wiest, O., & Zhang, X. (2024). Large language model based multi-agents: A survey of progress and challenges. In Proceedings of the 33rd International Joint Conference on Artificial Intelligence (IJCAI 2024) (pp. 8048–8057).

Hevner, A. R., March, S. T., Park, J., & Ram, S. (2004). Design science in information systems research. MIS Quarterly, 28(1), 75–105.

Hong, S., Zhuge, M., Chen, J., Zheng, X., Cheng, Y., Wang, J., Zhang, C., Wang, Z., Yau, S. K. S., Lin, Z., Zhou, L., Ran, C., Xiao, L., Wu, C., & Schmidhuber, J. (2024). MetaGPT: Meta programming for a multi-agent collaborative framework. In International Conference on Learning Representations (ICLR 2024). https://doi.org/10.48550/arXiv.2308.00352

Jonah, S., Moses, Q., Babatunde, A., Ajao-Olarinoye, M., & Bammeke, D. (2026). LLMs for agentic home energy management [Preprint]. arXiv:2607.04569. https://doi.org/10.48550/arXiv.2607.04569

Kert, K. (2023). Nordpool cheapest hours actions [Home Assistant blueprint]. GitHub. Retrieved 3 October 2026, from https://github.com/kullarkert/ha_nordpool_hours_blueprint

Lakens, D. (2017). Equivalence tests: A practical primer for t tests, correlations, and meta-analyses. Social Psychological and Personality Science, 8(4), 355–362. https://doi.org/10.1177/1948550617697177

Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W., Rocktäschel, T., Riedel, S., & Kiela, D. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. In Advances in Neural Information Processing Systems 33 (NeurIPS 2020) (pp. 9459–9474).

Li, B., Mellou, K., Zhang, B., Pathuri, J., & Menache, I. (2023). Large language models for supply chain optimization [Preprint]. arXiv:2307.03875. https://doi.org/10.48550/arXiv.2307.03875

Michelon, F., Zhou, Y., & Morstyn, T. (2025). Large language model interface for home energy management systems. In Proceedings of the 16th ACM International Conference on Future and Sustainable Energy Systems (e-Energy '25) (pp. 590–602). ACM. https://doi.org/10.1145/3679240.3734586

Murray, D., Stankovic, L., & Stankovic, V. (2017). An electrical load measurements dataset of United Kingdom households from a two-year longitudinal study. Scientific Data, 4, 160122. https://doi.org/10.1038/sdata.2016.122

Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., Shamseer, L., Tetzlaff, J. M., Akl, E. A., Brennan, S. E., Chou, R., Glanville, J., Grimshaw, J. M., Hróbjartsson, A., Lalu, M. M., Li, T., Loder, E. W., Mayo-Wilson, E., McDonald, S., ... Moher, D. (2021). The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. BMJ, 372, n71. https://doi.org/10.1136/bmj.n71

Peffers, K., Tuunanen, T., Rothenberger, M. A., & Chatterjee, S. (2007). A design science research methodology for information systems research. Journal of Management Information Systems, 24(3), 45–77.

Raghavan, V. S., & Giridhar, A. V. (2025). Cost-optimal residential energy scheduling via a zero-shot LLM agent and model predictive control. Journal of Building Performance Simulation, 19(3), 492–510. https://doi.org/10.1080/19401493.2025.2605465

Sou, K. C., Weimer, J., Sandberg, H., & Johansson, K. H. (2011). Scheduling smart home appliances using mixed integer linear programming. In Proceedings of the 50th IEEE Conference on Decision and Control and European Control Conference (pp. 5144–5149). IEEE. https://doi.org/10.1109/CDC.2011.6161081

Venable, J., Pries-Heje, J., & Baskerville, R. (2016). FEDS: A framework for evaluation in design science research. European Journal of Information Systems, 25(1), 77–89. https://doi.org/10.1057/ejis.2014.36

Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., & Cao, Y. (2023). ReAct: Synergizing reasoning and acting in language models. International Conference on Learning Representations (ICLR 2023). arXiv:2210.03629.